# Given a real GitHub issue and its code, how often can a judge correctly tell the actual fix from a change that only looks like one?

15 judges, the same 100 test-verified issues and the same 938 candidate changes each. **Headline: balanced accuracy** = average of “real fixes accepted” (of 200) and “wrong changes rejected” (of 738).

| # | JUDGE | SIZE parameters | BALANCED ACCURACY headline | REAL FIXES ACCEPTED of 200 | WRONG CHANGES REJECTED of 738 |
|---|---|---|---|---|---|
| 1 | Claude Opus 5 | undisclosed · est. 5–10 T (MoE) | 86.0 % | 191 | 565 |
| 2 | GPT-5.6 Sol | undisclosed · est. 2–5 T | 83.5 % | 178 | 576 |
| 3 | Coding World Model Predictor (Forge) | 137 M | 80.9 % | 191 | 490 |
| 3 | Gemini 3.8 Flash | undisclosed | 80.9 % | 182 | 523 |
| 5 | gpt-oss-20b | 21 B MoE · 3.6 B active | 78.6 % | 159 | 573 |
| 6 | Gemma 4 12B | 12 B · 6-bit, local | 77.1 % | 186 | 452 |
| 7 | Qwen3.8-27B | 27.8 B · 4-bit, local | 76.5 % | 192 | 421 |
| 8 | DeepSeek V3.2 | 671 B MoE · 37 B active | 74.6 % | 195 | 381 |
| 9 | Qwen3-32B | 32.8 B | 73.8 % | 194 | 373 |
| 10 | Qwen3-14B | 14.8 B · 4-bit, local | 73.0 % | 162 | 480 |
| 11 | Llama 3.3 70B | 70 B | 72.1 % | 191 | 359 |
| 12 | Gemma 3 27B | 27 B | 63.4 % | 186 | 249 |
| 13 | Phi-4 | 14 B · 4-bit, local | 57.2 % | 175 | 199 |
| 14 | Qwen2.5-Coder 7B | 7.6 B · 4-bit, local | 57.1 % | 60 | 621 |
| 15 | Mistral-Nemo 12B | 12.2 B · 4-bit, local | 20.1 % | 27 | 197 |

## WHAT THIS TEST MEASURES

Code understanding without any external test. The judge sees only three things: the issue written in plain language, the code as it was, and a proposed change. To score the change it has to relate the goal stated in words to the behaviour of the written code and predict the consequence of the change: whether the program, after this change, does what the issue asks. There is no test suite to run, no execution, no hint about which candidate is real. The ground truth comes from real fixes whose test results are known, and the judge never sees those results.

In world model terms the score is the accuracy of the judge's prediction of what a change does to a program. A judge that scores real fixes near zero and broken or unrelated changes near one is predicting consequences correctly. That prediction is what a coding world model exists to make, and it is the capability this benchmark isolates.

## THE TEST

100 issues drawn from 813 test-verified GitHub fixes (SWE-bench-family sources, held out by repository from all training data). For each issue the judge scores 9–10 candidate changes one at a time: the real fix, a cosmetically altered copy of it, up to four broken versions of it, up to two unrelated commits from the same repository, a fix for a different issue, and an empty change. Input per call: the issue text, the code before the change, and the change itself, up to 8,192 tokens, byte-identical for every judge.

Score scale: 0 = exactly the fix, 100 = not the fix. A decision is right when a real fix scores below 50 and a wrong change above 50.

## HOW TO READ IT

The Predictor (137 M parameters) accepts real fixes as reliably as the best model and rejects broken versions of the fix better than any of the 14 LLMs (0.62 mean score vs 0.60 for GPT-5.6). Its remaining gap is changes that are valid but for a different issue.

Sizes for open models are as reported by their authors; the three closed models are undisclosed and the figures shown are third-party estimates. The Predictor and Gemini 3.8 Flash tie at the third decimal. With 100 issues, differences under about 3 points are within sampling noise. LLMs were called with deterministic settings through their vendor APIs (Bedrock, OpenAI, Gemini) or run locally at the quantisation shown.

## COST OF OPERATION

Cost of one full run of this benchmark per judge: 938 calls, vendor list prices, retries included, computed from the logged token counts. Local models and the Predictor run on a single workstation GPU (one RTX 4090) and carry no API cost.

| JUDGE | RUN COST | WHERE IT RAN |
|---|---|---|
| Claude Opus 5 | $57 | API (AWS Bedrock) |
| GPT-5.6 Sol | $38 | API (OpenAI) |
| Gemini 3.8 Flash | $6 | API (Google) |
| Llama 3.3 70B | $5 | API (AWS Bedrock) |
| DeepSeek V3.2 | $4 | API (AWS Bedrock) |
| Qwen3-32B, gpt-oss-20b, Gemma 3 27B | $1 each | API (AWS Bedrock) |
| Coding World Model Predictor (Forge), 137 M | $0 | local GPU, 7,456 pairs in 4 minutes |
| Gemma 4 12B, Qwen3.8-27B, Qwen3-14B, Phi-4, Qwen2.5-Coder 7B, Mistral-Nemo 12B | $0 | local GPU (llama.cpp) |